



GIF Signature Format: Introduction & Recovery

Graphics Interchange Format

Type code	GIF
File extension	.gif
Type of format	Raster/lossless
Colors	up to 8 bits
Compression	LZW (Lempei-Ziv-Welch)
Numerical Format	Little-endian
Magic number (GIF87a)	47 49 46 38 37 61
Magic number (GIF89a)	47 49 46 38 39 61
Developer	CompuServe
Release Date	1987

Developed by CompuServe in 1987, a bitmap image format known as GIF (Graphics Interchange Format has since received widespread use on the internet due to its wide support and portability.

GIF format supports up to 8 bits per pixel for each image, allowing a single image to reference its palette of up to 256 different colors chosen from the 24-bit RGB color space.

It supports animations and allows a separate palette of up to 256 colors for each frame. GIF images are compressed using the LZW lossless data compression technique to reduce the file size without degrading the visual quality of the image.

GIF files start with a fixed-length header (GIF87a or GIF89a) where 7a and 9a specify the version, followed by a fixed-length Logical Screen Descriptor giving the size and other characteristics of the logical display. Screen Descriptor may also specify the presence & size of a Global Color Table, which follows next (if present).

After that, the file is divided into segments, each introduced by a 1-byte prefix:

- An image - introduced by hex value: 2C, a comma sign (,)
- An extension block - introduced by hex value: 21, an exclamation point (!)
- The trailer - a single byte of hex value: 3B, a semi-colon (;), which should be the last byte of the file

An image starts with a fixed-length Image Descriptor, which may specify the presence and size of a Local Color Table (which follows next if present).

The code is being continued with image data: one byte giving the bit width of the unencoded symbols (which must be at least 2 bits wide, even for bi-color images).

It is later followed by a linked list of sub-blocks containing the LZW-encoded data.

Extension blocks (blocks that "extend" the 87a definition via a mechanism already defined in the 87a spec) consist of the prefix, an additional byte specifying the type of an extension, and a linked list of sub-blocks with the extension data.

Extension blocks that modify an image (like the Graphic Control Extension that specifies the optional animation delay time and optional transparent background color) must immediately precede the segment with the image to which they are referring. The linked lists used by the image data and the extension blocks consist of series of sub-blocks, each sub-block beginning with a byte giving the number of subsequent data bytes in the sub-block (1 to 255). The series of sub-blocks is terminated by an empty sub-block (a 0 byte). A closer look at the GIF image



When inspecting sample.gif file's binary data using any Hex Viewer, like Active@ Disk Editor we can see it starts with a signature GIF8 (hex: 47 49 46 38). It is followed with the version 9a (hex: 39 61), and then goes logical screen width: 280 pixels (hex 18 01) and height 90 pixels (hex: 5A 00). All multi-byte values in GIF structures are in little-endian order (lowest byte goes first).

Offset	0	1	2	3	4	5	6	7	-	8	9	A	B	C	D	E	F	ASCII
00000000	47	49	46	38	39	61	18	01		5A	00	E6	00	00	FF	FF	FF	GIF89a..Z.ж..яяя
00000010	F7	E2	51	EA	D8	58	D9	B4		0B	00	00	00	F5	E0	50	E8	чвQкШХЩГ....хари
00000020	D6	57	FA	E1	4D	4C	F9	3F		4C	F9	3F	54	D5	B9			ЦWъштпШМК°ФвОТХ№
00000030	40	EA	D2	41	E5	40	EA	D2		40	EA	D2	41	E5	C9	9C		@кТКЮЙQeTVЭГЕШЙнь
00000040	DD	CC	9A	D4	BC	4B	F2	DC		DD	CC	9A	D4	BC	4B	F2	DC	ЭМЬфjКтЬOrKGД\$Хе
00000050	C7	2B	D9	BD	42	BD	A1	5C		F3	D5	26	DC	C7	50	DF	CB	З+ЩSBSŸ\yX&ЪЭРЯЛ
00000060	52	ED	D5	4C	E9	D7	58	DD		C5	64	E7	BC	0A	FE	D5	00	РНХLЙчХЭEdэj.юX.
00000070	E0	CC	52	D3	B6	40	D7	C4		8F	C6	A5	39	B5	97	3A	F6	амРУѠ@чДЦЖГ9μ-:ц
00000080	E2	24	FF	FE	00	CD	B4	48		D1	B3	3F	FF	E9	00	BF	9E	в\$яю.НгHci?яй.іћ
00000090	36	D4	BF	69	F6	EE	24	D9		C2	4E	D6	B2	18	F7	F3	E6	6Фiіцo\$ЩBNЦI.чуж
000000A0	BB	9E	3E	ED	DE	97	D5	C0		69	DB	BE	36	E3	D3	9C	D7	»ћ>нЮ-ХAiNs6гУньЧ
000000B0	B6	35	DD	CA	6E	D9	BE	3C		E8	CE	08	E6	D9	9A	E7	D2	Ѡ5ЭKnЩs<иО.жЩъЭТ
000000C0	69	DB	C1	32	C1	9F	36	B7		98	3B	DE	CF	9F	D4	B0	25	иЫБ2Вц6..;ЮПцФ°%
000000D0	00	00	00	00	00	00	00	00		00	00	00	00	00	00	00	00	
000000E0	00	00	00	00	00	00	00	00		00	00	00	00	00	00	00	00	
000000F0	00	00	00	00	00	00	00	00		00	00	00	00	00	00	00	00	
00000100	00	00	00	00	00	00	00	00		00	00	00	00	00	00	00	00	
00000110	00	00	00	00	00	00	00	00		00	00	00	00	00	00	00	00	
00000120	00	00	00	00	00	00	00	00		00	00	00	00	00	00	00	00	
00000130	00	00	00	00	00	00	00	00		00	00	00	00	00	00	00	00	
00000140	00	00	00	00	00	00	00	00		00	00	00	00	00	00	00	00	
00000150	00	00	00	00	00	00	00	00		00	00	00	00	00	00	00	00	
00000160	00	00	00	00	00	00	00	00		00	00	00	00	00	00	00	00	
00000170	00	00	00	00	00	00	00	00		00	00	00	00	00	00	00	00	
00000180	00	00	00	00	00	00	00	00		00	00	00	00	00	21	F9	04	!щ.
00000190	00	00	00	00	00	2C	00	00		00	00	18	01	5A	00	00	07	Z...
000001A0	FF	80	00	82	83	84	85	86		87	88	89	8A	8B	8C	8D	8E	т€%Љ<ЬќѠ
000001B0	8F	90	91	92	93	94	95	96		97	98	99	9A	9B	9C	9D	9E	ѠЉ>Ьќћ

Flags byte at offset 0A (hex) is E6 (hex), high bit is set, which shows that Global Color Table is following. Its size is calculated as:

$$(1 \ll ((0xE6 \& 0x07) + 1)) * 3 = 384 \text{ bytes.}$$

Add 13 (GIF header size) to 384 (GCT size) and the result is 397. Thus at offset 397 (hex: 018D) first data block starts. The first block starts with an exclamation sign (! or 21 hex) which means it is an extension block.

Extension block size is fixed, and it is 8 bytes. Add 8 bytes to 397, and go to the next block at offset 405 (hex: 0195). The second block starts with a comma sign (, or 2C hex) which means it is an image block. It has fixed size header (11 bytes). Image header's last byte (at offset 019F hex) is a GCT flag: 07.

Calculating GCT size the same way as in GIF primary header. A high bit (hex: 80) is NOT set, so no GCT is followed and no calculations for color table required. After image header, the next in line are sub-blocks. First sub-block has its size in a first byte, which is 255 (hex value FF at offset: 01A0 hex).

Next sub-block offset is 672 (405 + 11 + 1 + 255). Keep iterating blocks and sub-blocks. This sample has 4 sub-blocks by 255 bytes each, at offsets: 01A0, 02A0, 03A0, 04A0 hex, and one sub-block 126 bytes (hex: 7E) at offset 05A0 hex. At offset 061F hex (05A0+7E hex) there is a zero byte 00 which says that no more sub-blocks in the current block. Moving to the next byte (offset 0620 hex). The next block starts with a semi-colon (; or 3B hex), which points to the end of the file. Thus, this GIF image file size is 1,569 bytes (hex: 0620+1).

Like

Share

2.3K people like this. Be the first of your friends.

About

LSoft Technologies Inc. is a privately owned North American software company. Our goal is to create world’s leading data recovery, security and backup solutions by providing rock solid performance, innovation, and unparalleled customer service.

Company

- About Us
- Partners
- Contact Us
- Privacy & Terms

Contact Us

sales @ lsoft.net
+1 (877) 403-8082
+1 (905) 812-8434
9am to 6pm EST

  

Address

LSoft Technologies Inc.
7177 Danton Promenade
Mississauga, ON
L5N 5P3
Canada